

# Postal MS

## Time Complexity Analysis

CST2550 | The Leftovers | Middlesex University

### 1. Introduction

This document provides a detailed time complexity analysis of all three custom data structures implemented in the PostalMS Postal Service Management System. All three structures were built from scratch without using any standard template library collections, as required by the CST2550 coursework brief.

Time complexity describes how the execution time of an algorithm grows as the size of the input increases. It is expressed using Big O notation. For PostalMS, understanding time complexity is critical because the system must handle large numbers of parcels and deliveries efficiently.

#### 1.1 Big O Notation Key

- $O(1)$  — Constant time. The operation takes the same time regardless of how many parcels are in the system. This is the best possible complexity.
- $O(\log n)$  — Logarithmic time. The operation time grows slowly as parcels increase. Each step eliminates half the remaining options.
- $O(n)$  — Linear time. The operation time grows proportionally with the number of parcels. Acceptable for full traversals.
- $O(n \log n)$  — Linearithmic time. Typical for sorting algorithms. Worse than  $O(n)$  but better than  $O(n^2)$ .

#### 1.2 Why Time Complexity Matters for PostalMS

PostalMS is designed to handle a growing number of parcels over time. A postal service could have thousands or even millions of parcels in its system. If the search operation was  $O(n)$  (linear), searching for one parcel out of one million would require checking up to one million records. With the Hash Table providing  $O(1)$  average search, the lookup time stays constant regardless of how many parcels exist.

## 2. Custom Hash Table — Time Complexity

### 2.1 Overview

The CustomHashTable converts a parcel tracking ID into an array index using a polynomial rolling hash function. This allows direct access to the stored value without searching. The average case for all core operations is  $O(1)$  because the hash function calculates the exact position of any key instantly.

### 2.2 How the Hash Function Affects Complexity

The hash function runs in  $O(k)$  time where  $k$  is the length of the tracking ID string. Because all PostalMS tracking IDs have the same fixed format PS-YYYY-NNNNN (14 characters),  $k$  is always constant. This means the hash function itself is effectively  $O(1)$  for this system.

### 2.3 When Worst Case Occurs

The worst case of  $O(n)$  occurs when all  $n$  tracking IDs hash to the exact same index, creating a single collision chain of length  $n$ . Every search, add and delete would then need to traverse the entire chain. In practice this is extremely unlikely with a good hash function and the auto-resize mechanism keeping the load factor below 0.75.

### 2.4 Auto-Resize Impact

When the load factor exceeds 0.75, the table doubles in size and re-inserts all existing items. This resize is  $O(n)$  but it happens rarely, approximately once every time the table fills up 75% of its capacity. The amortized cost of resize spread across all insertions is  $O(1)$  per insertion, so the overall Add operation remains  $O(1)$  amortized.

### 2.5 Time Complexity Table

| Operation | Best Case | Average | Worst Case | Reason for Worst Case                                      |
|-----------|-----------|---------|------------|------------------------------------------------------------|
| Add       | $O(1)$    | $O(1)$  | $O(n)$     | All keys hash to same slot – full chain traversal required |
| Search    | $O(1)$    | $O(1)$  | $O(n)$     | Target key is at the end of a very long collision chain    |
| Delete    | $O(1)$    | $O(1)$  | $O(n)$     | Target key is at the end of a very long collision chain    |
| Resize    | $O(n)$    | $O(n)$  | $O(n)$     | Every existing item must be re-hashed into the new table   |
| Clear     | $O(n)$    | $O(n)$  | $O(n)$     | Every slot in the array must be individually set to null   |

## 3. Custom Binary Search Tree — Time Complexity

### 3.1 Overview

The CustomBST stores parcels in a tree where every node's left subtree contains smaller tracking IDs and every node's right subtree contains larger tracking IDs. This ordering property allows the search to eliminate half the remaining nodes at each step, giving  $O(\log n)$  average performance.

### 3.2 What $O(\log n)$ Means in Practice

With  $O(\log n)$  search, a BST containing 1000 parcels requires at most 10 comparisons to find any parcel. A BST with 1 million parcels requires at most 20 comparisons. Each time the number of parcels doubles, only one extra comparison is needed. This is significantly better than  $O(n)$  linear search which would require up to 1 million comparisons for 1 million parcels.

### 3.3 When Worst Case Occurs — Degenerate Tree

The worst case of  $O(n)$  occurs when all tracking IDs are inserted in already sorted order. For example, inserting PS-2026-00001 then PS-2026-00002 then PS-2026-00003 in sequence creates a degenerate tree where every node only has a right child. The tree degrades into a linked list and search becomes  $O(n)$ . In PostalMS this is mitigated because tracking IDs are assigned sequentially but parcels from different users are interleaved, preventing a perfectly sorted insertion order.

### 3.4 In-Order Traversal is Always $O(n)$

In-order traversal always visits every node exactly once regardless of the tree structure. This is intentional the purpose of in-order traversal is to produce a complete sorted list of all parcels, so visiting every node is required.  $O(n)$  is the best possible complexity for any algorithm that must process every item.

### 3.5 Time Complexity Table

| Operation | Best Case   | Average     | Worst Case | Reason for Worst Case                                      |
|-----------|-------------|-------------|------------|------------------------------------------------------------|
| Insert    | $O(\log n)$ | $O(\log n)$ | $O(n)$     | Sorted input creates a degenerate tree (linked list shape) |
| Search    | $O(1)$      | $O(\log n)$ | $O(n)$     | Target is at deepest leaf of a degenerate tree             |
| Delete    | $O(\log n)$ | $O(\log n)$ | $O(n)$     | Target is at deepest leaf of a degenerate tree             |
| InOrder   | $O(n)$      | $O(n)$      | $O(n)$     | Must visit every single node exactly once                  |

| Operation | Best Case | Average     | Worst Case | Reason for Worst Case                                        |
|-----------|-----------|-------------|------------|--------------------------------------------------------------|
| FindMin   | $O(1)$    | $O(\log n)$ | $O(n)$     | Must follow left child pointers to the deepest left leaf     |
| Clear     | $O(1)$    | $O(1)$      | $O(1)$     | Just sets root to null – garbage collection handles the rest |

## 4. Custom Queue — Time Complexity

### 4.1 Overview

The CustomQueue is implemented as a singly linked list with two pointers: front points to the first node for dequeue operations and rear points to the last node for enqueue operations. Because both pointers provide direct access to their respective ends, all core operations are  $O(1)$  constant time with no exceptions.

### 4.2 Why the Queue has No Worst-Case Degradation

Unlike the Hash Table and BST, the Queue has no worst-case scenario for its core operations. Enqueue always adds to the rear pointer in  $O(1)$  regardless of queue size. Dequeue always removes from the front pointer in  $O(1)$  regardless of queue size. There are no chains to traverse, no tree branches to follow, and no hash collisions to handle. The Queue is the most predictable data structure in PostalMS.

### 4.3 ToList is $O(n)$ by Design

The ToList operation traverses every node from front to rear to build a complete list of all pending deliveries. This is  $O(n)$  because it must visit every node. This is intentional and acceptable because the purpose of ToList is to return all items, so processing all  $n$  items is the required behaviour.

### 4.4 Time Complexity Table

| Operation | Best Case | Average | Worst Case | Reason for Worst Case                                          |
|-----------|-----------|---------|------------|----------------------------------------------------------------|
| Enqueue   | $O(1)$    | $O(1)$  | $O(1)$     | Rear pointer gives direct access to last node – no traversal   |
| Dequeue   | $O(1)$    | $O(1)$  | $O(1)$     | Front pointer gives direct access to first node – no traversal |

| Operation | Best Case | Average | Worst Case | Reason for Worst Case                           |
|-----------|-----------|---------|------------|-------------------------------------------------|
| Peek      | $O(1)$    | $O(1)$  | $O(1)$     | Simply reads the value at the front pointer     |
| IsEmpty   | $O(1)$    | $O(1)$  | $O(1)$     | Simply checks whether the front pointer is null |
| ToList    | $O(n)$    | $O(n)$  | $O(n)$     | Must traverse every node from front to rear     |
| Clear     | $O(1)$    | $O(1)$  | $O(1)$     | Just sets front and rear pointers to null       |

## 5. Overall Comparison

### 5.1 Side by Side Comparison

The table below compares the average case time complexity of all three data structures across common operations. Green indicates the best performer for each operation.

| Operation     | Hash Table    | BST         | Queue  | Notes                                               |
|---------------|---------------|-------------|--------|-----------------------------------------------------|
| Add / Insert  | $O(1)$        | $O(\log n)$ | $O(1)$ | Hash Table is fastest for insertion on average      |
| Search        | $O(1)$        | $O(\log n)$ | $O(1)$ | Hash Table is fastest, Queue has no search          |
| Delete        | $O(1)$        | $O(\log n)$ | $O(1)$ | Hash Table is fastest for deletion on average       |
| Sorted Output | $O(n \log n)$ | $O(n)$      | N/A    | BST in-order gives sorted output without re-sorting |
| FIFO Order    | N/A           | N/A         | $O(1)$ | Queue is the only structure with guaranteed FIFO    |
| Worst Case    | $O(n)$        | $O(n)$      | $O(1)$ | Queue has no worst case degradation                 |

### 5.2 Why Each Structure Was Chosen

Each data structure was selected because it is the optimal choice for its specific role in PostalMS:

#### Hash Table — Chosen for Parcel Search

When a user enters a tracking ID, the system needs the fastest possible lookup. The Hash Table provides  $O(1)$  average search which no other structure can match for this

use case. A sorted array would give  $O(\log n)$  binary search. A linked list would give  $O(n)$  linear search. The Hash Table is the clear winner for lookup speed.

### **BST — Chosen for Sorted Dashboard Display**

The parcel dashboard needs to display all parcels sorted by tracking ID. The BST maintains sorted order automatically as items are inserted, so an in-order traversal produces a sorted list in  $O(n)$  without any additional sorting step. If we used an unsorted array instead, we would need  $O(n \log n)$  to sort it every time the dashboard loads. The BST eliminates this overhead entirely.

### **Queue — Chosen for Delivery Processing**

Deliveries must be processed in the exact order they were assigned. The Queue is the only data structure that naturally enforces FIFO ordering with  $O(1)$  enqueue and dequeue. A sorted list would require  $O(n)$  insertion to maintain order. A stack would give LIFO (last in first out) which is the opposite of what is needed. The Queue is the only correct choice for this requirement.

## **5.3 Space Complexity**

All three data structures use  $O(n)$  space where  $n$  is the number of items stored. Each item requires one node object containing the key, value and pointer fields. The Hash Table additionally uses  $O(m)$  space for the array of size  $m$ , but since  $m$  is proportional to  $n$  (kept below load factor 0.75), the overall space complexity remains  $O(n)$ .

## **6. Conclusion**

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PostalMS uses three complementary custom data structures each chosen for its optimal time complexity characteristics in its specific role. The Hash Table provides  $O(1)$  average parcel lookup which is critical for a responsive user experience. The BST provides  $O(\log n)$  insertion and  $O(n)$  sorted traversal which is optimal for the dashboard display. The Queue provides  $O(1)$  enqueue and dequeue with guaranteed FIFO ordering which is essential for fair delivery processing.

All three structures were implemented from scratch without using any standard template library collections. The implementations have been verified through 59 unit tests all of which passed successfully confirming that the time complexity characteristics described in this document are correctly implemented in the PostalMS codebase.